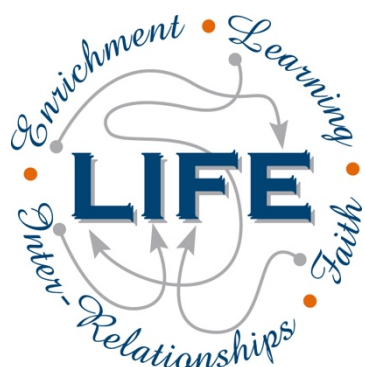




Holy Cross College

**Semester 1 Examination 2019
Question/Answer Booklet**

**Year 10 Earth and Space
Science**

Please place your student identification label in this box

Student Name _____

Student's Teacher _____

Article I. Time allowed for this paper

Working time for paper: 30 minutes

Materials required/recommended for this paper

Article II. To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	10	10	10	10	40
Section Two: Short Answer	8	8	20	21	60
				31	100

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer Booklet.
3. Working or reasoning should be clearly shown when calculating or estimating answers.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Section I: Multiple choice questions (circle the correct answer)

1. Astronomers today believe the Universe has:

- no centre and that even the most powerful telescopes cannot see all of it.
- a centre and that even the most powerful telescopes cannot see all of it.
- no centre and that the most powerful telescopes can easily see all of it.
- a centre and that the most powerful telescopes can easily see all of it.

[1 mark]

2. Most stars are made almost entirely of:

- hydrogen and helium.
- sulfur and oxygen.
- hydrogen and potassium.
- helium and carbon dioxide.

[1 mark]

3. Approximately how many years ago is our solar system thought to have been formed?

- 5 billion
- 5 trillion
- 5 thousand
- 5 million

[1 mark]

4. Cosmic microwave background (CMB) radiation is evidence for the:

- Big Bang theory of the Universe.
- cyclic theory of the Universe.
- big crunch theory of the Universe.
- Steady State theory of the Universe.

[1 mark]

5. A star is 6.62×10^{14} km away from Earth. Approximately how many light years distance is this if one light year is 9.46×10^{12} km?

- 700
- 7000
- 70
- 7

[1 mark]

6. The more red a star is, the :

- cooler it is.
- hotter it is.
- smaller it is.
- larger it is.

[1 mark]

7. The space between stars is filled with dust and gas called:

- nebulae.
- galaxies.
- constellations.
- supernovae.

[1 mark]

8. When a star that is close to Earth is observed in April, and then six months later in October, its apparent position against the background of distant stars appears to change. This is known as:

- stellar polaroid.
- stellar brightness.
- stellar parallax.
- stellar movement.

[1 mark]

9. The change in wavelength of sound waves emitted from a moving sound source is known as:

- the Doppler effect.
- red shift.
- Einstein's equation.
- Hubble's Law.

[1 mark]

10. The conclusion from Hubble's Law is that:

- the more distant galaxies are travelling the fastest.
- we are at the centre of the Universe.
- everything is moving away from Earth.
- the further a galaxy is away from Earth, the slower its speed.

[1 mark]

Section II: Short answer questions

- 11.** Using the data below, calculate how many seconds light would take to reach Earth from the Sun.

speed of light = 300 000 km/s

distance to Sun = 150 000 000 km

[2 mark]

- 12.** What does the Big Bang theory state?

[2 mark]

- 13.** Two stars in the night sky appear to be the same brightness. Are they the same distance from Earth? Explain your answer.

[2 mark]

- 14.** The closest star outside of our own solar system is Proxima Centauri, at 4.2 light years away. Explain what this distance means.

[2 mark]

-
- 15.** How can a weak dim torch appear brighter than a new strong torch? Explain the relevance of this comparison to stars.

[3 mark]

- 16.** Explain exactly what is occurring inside a main sequence star to produce energy.

[3 mark]

- 17.** Arrange the following stages in the life cycle of a star in the correct order from earliest to latest:

red giant, white dwarf, main sequence star, black dwarf

[4 mark]

- 18.** A racing car makes a ‘yeee ... owww’ sound as it speeds towards and then past an observer. Choose the correct terms in the following statement, relating the sound to the blue and red shift of the Doppler effect and to the movement of galaxies towards and away from us.

The ‘yeee’ is a higher/lower frequency, which corresponds to a blue/red shift in light, which indicates movement towards/away from us.

[3 mark]